THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Start with $n = 26$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 10 **(b)** 11 **(c)** 12 **(d)** 9

2. Using Euclid's subtraction algorithm for gcd(1344, 872), the number of subtraction steps needed is: 1

(a) 16 **(b)** 15 **(c)** 13 **(d)** 14

3. Using the division form of Euclid's algorithm for gcd(2505, 898), the number of divisions needed is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 8

4. Set r = 0. While n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Starting with n = 5702, the final r is: 1

(a) 2075 **(b)** 2084 **(c)** 2066 **(d)** 2165

5. Using gcd(82, 298) = 2, the LCM of 82 and 298 is: 1

(a) 12300 **(b)** 12220 **(c)** 12218 **(d)** 12216

6. To test whether 2381 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{2381}\rfloor$. The last number checked is: 1

(a) 49 **(b)** 47 **(c)** 1190 **(d)** 48

7. A counter starts at 0. For i = 1, 2, …, 46, the counter increases by 1 whenever i is divisible by 3 or by 5. The final counter value is: 1

(a) 23 **(b)** 22 **(c)** 20 **(d)** 21

8. By Euclid's algorithm, gcd(4221, 3899) is: 1

(a) 9 **(b)** 8 **(c)** 6 **(d)** 7

9. Start with $n = 23$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 17 **(b)** 16 **(c)** 14 **(d)** 15

10. Using Euclid's subtraction algorithm for gcd(1971, 1251), the number of subtraction steps needed is: 1

(a) 14 **(b)** 13 **(c)** 16 **(d)** 15

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the gcd of 320, 294 and 284 by applying Euclid's division algorithm twice: first to two numbers and then to the result and the third. 2

12. A floor of 429 cm by 220 cm is to be covered by identical square tiles, as large as possible, with no cutting. Use Euclid's algorithm to find the tile side and the number of tiles. 2

13. Find the gcd of 346 and 350 by the division form of Euclid's algorithm and use it to find their LCM. 2

14. Find the gcd of 844, 854 and 358 by applying Euclid's division algorithm twice: first to two numbers and then to the result and the third. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Trace: count = 0; for i = 1 to 11: for j = i to 11: count = count + 1. Find the final count and explain the pattern. 3

16. Trace: total = 0; for i = 1 to 11: if i is a multiple of 3, add i^2 to total. List the numbers added and the final total. 3

17. Start with n = 23. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

OR

Trace: count = 0; for i = 1 to 6: for j = i to 6: count = count + 1. Find the final count and explain the pattern. 3

18. Trace: total = 0; for i = 1 to 8: if i is a multiple of 4, add i^2 to total. List the numbers added and the final total. 3

OR

Start with n = 20. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the largest number that divides both 2028 and 1116 exactly, using the division algorithm. Hence list all the common divisors of the two numbers. 5

OR

Find the gcd of 621, 426 and 1602 by applying Euclid's division algorithm twice: first to two numbers and then to the result and the third. 5

20. Find the largest number that divides both 2175 and 975 exactly, using the division algorithm. Hence list all the common divisors of the two numbers. 5

OR

Find the gcd of 2196, 1936 and 1236 by applying Euclid's division algorithm twice: first to two numbers and then to the result and the third. 5

END OF QUESTION PAPER — Total: 40 Marks